

Sergio A. Barbosa

MENG IN COMPUTER SCIENCE AND SYSTEMS ENGINEERING & MSc IN ARTIFICIAL INTELLIGENCE

Bogotá, Colombia

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Profile

I am passionate about intelligent systems and the intersection of AI and Quantum Computing. My research focuses on leveraging quantum principles to enhance AI, particularly in conversational agents and NLP. With a background in Computer Science, I have worked on AI-driven systems applied to oncology, business, and social impact, aiming to address challenges with high societal impact.

Education

Pontificia Universidad Javeriana

MENG IN COMPUTER SCIENCE AND SYSTEMS ENGINEERING

- Degree with honours, *Magna Cum Laude*. Recognition of Academic Excellence (GPA 4.7/5.0).

Bogotá, Colombia

Feb. 2024 - Nov. 2024

Pontificia Universidad Javeriana

MSC IN ARTIFICIAL INTELLIGENCE

- Degree with honours, *Cum Laude*. Recognition of Academic Excellence (GPA 4.6/5.0).

Bogotá, Colombia

Jul. 2021 - Dec. 2022

The Naval School 'Almirante Padilla'

B.S. IN ELECTRONIC ENGINEER

- Degree with honours. Scholarship for excellent academic results (GPA 9.11/10).

Cartagena, Colombia

Jan. 2014 - Dec. 2018

Work Experience

Digital GHENOVA

ENGINEER OF AI/NLP

- Expertly developed innovative technological solutions based on language models for the creation of conversational agents for digital transformation, applied in areas such as healthcare and business. My skills include AI generative techniques, NLP, fine-tuning Large Language Models (LLMs), designing conversational agents, applying DL and ML techniques, OCR applications, dialog management, and speaker diarisation.

Bogotá, Colombia

Jul. 2022 - Present

Ministry of Information Technologies and Communications

ENGINEER OF AI/NLP (CONTRACTOR)

- Develop and implement AI and NLP algorithms to assess data quality, establishing quality criteria for the open data published on the portal of the Ministry of Information Technologies and Communications of Colombia. Skills include working with language large models, deep learning and data science.

Bogotá, Colombia

Oct. 2024 - Dec. 2024

(JEP) Special Jurisdiction for Peace Organisation

ENGINEER OF NLP (CONTRACTOR)

- Successfully adapted and deployed a Named Entity Recognition (NER) model pre-trained in the social context of victims and the armed conflict in Colombia, improving the accuracy and understanding of narratives surrounding this complex socio-political landscape. Skills include working with pre-trained NER models, Transformer Language Models (TLMs), integrated DL models, implementing ML models, and data science.

Bogotá, Colombia

Sep. 2023 - Dec. 2023

(COTECMAR) Science and Technology Corporation

ENGINEER & NAVAL OFFICER

- Member of the Colombian Navy, contributing to the production team for the construction of Colombia's first Marine Scientific Research Vessel and other maritime patrol vessels. Naval Officer in service since 2011. Skills include leadership, problem-solving, resource optimisation, management, and process improvement.

Cartagena, Colombia

Jan. 2016 - Dec. 2021

Publications

Detection of Hate Speech, Racism and Misogyny in Digital Social Networks: Colombian Case Study

ACADEMIC PUBLICATION. BIG DATA AND COGNITIVE COMPUTING. 8, 113. [HTTPS://DOI.ORG/10.3390/bdcc8090113](https://doi.org/10.3390/bdcc8090113)

- The main contribution of this work is the fine-tuning of a model using a proprietary dataset extracted from social networks, georeferenced in Colombia, and manually labelled with typical and dialect-specific features. Indeed, the experiment compares the performance of TLM models adjusted to the Colombian slang domain against a GPT, having a significant impact on achieving more accurate predictions in this task.

MDPI Journal

Sept. 2024

- This study explores how mood manifests in text publications on Twitter, aiming to identify patterns indicative of depressive, anxious, and potentially suicidal language. This work involves training models using Word Embeddings methodology and applying Latent Dirichlet Allocation (LDA) to group topics, comparing their performance with LLMs.

Honours & Awards

2025	Magna Cum Laude , Degree with honours in Master’s in Computer Science.	Bogotá, Colombia
2025	Meritorious thesis in Master’s in Computer Science , Designed and implemented a modular and scalable architecture for a conversational agent focused on querying the Truth Commission’s legacy, using a multilingual Retrieval-Augmented Generation (RAG) system with Large Language Models for precise searches in both English and Spanish.	Bogotá, Colombia
2023	Cum Laude , Degree with honours in Master’s in Artificial Intelligence.	Bogotá, Colombia
2022	Meritorious thesis in Master’s in Artificial Intelligence , in computer vision achieved by implementing an AI model utilising Reinforcement Learning (RL) and Convolutional Neural Network (CNN) to label students’ emotions during virtual classes, based on their facial features, gaze, and gestures.	Bogotá, Colombia
2022	Academic Distinctions , Recognitions of Academic Excellence (3x).	Bogotá, Colombia
2016	Medal “Francisco José de Caldas” category “to consecration” , Awarded for having obtained the best grade-point average in the academic program of Electronic Engineering.	Cartagena, Colombia
2018	Order of Naval Merit “Almirante Padilla” category “Officer” , For military virtues, dedication to work, distinguished professional services, exemplary discipline.	Cartagena, Colombia

Research Experience

2024 CFP Tech Ethics Lab, University of Notre Dam	Bogotá, Colombia
TEAM MEMBER OF PROJECT	Jun 2024 — Dec. 2024
• The goal of this project is to create a Question-Answering System for the Colombian Truth Commission, the first-ever digital archive of the peace process. I am part of the research team at Pontificia Universidad Javeriana (PUJ) and participate in the development of the project.	
National Patent P202331094	Seville, Spain
TEAM INVENTOR, GHENOVA DIGITAL SLU	Jul. 2022 - Oct. 2024
• Publication No. ES2968547: System for Implementing a Digital Navigation Notebook Managed by Voice and Integrated with the Data Acquisition System of a Vessel, and Application Procedure.	
Session class in Data Analytics Program at Pontificia Universidad Javeriana	Bogotá, Colombia
LECTURER	Oct. 2023
• Predictive analytical methods of Machine Learning: decision trees, regression, ensemble models.	
Center of Excellence in Big Data & Data Analytics	Bogotá, Colombia
DEVELOP TEAM OF PROJECT	Jul 2022 — Dec 2022
• Successfully development of research project about text classification in the social context. Course advanced topics of AI.	

Courses & Certificates

2025	Quantum Computing in Practice , IBM Quantum Learning (Online)	Currently in progress
2025	Software Design and Architecture Specialisation , University of Alberta.	Online
2025	The Complete Quantum Computing Course , Udemy	Online
2024	Introduction to Quantum Computing , University College London (UCL).	Online
2023	6.86x: Machine Learning with Python-From Linear Models to Deep Learning , MITx - EdX	Online
2022	SQL for Data Science , University of California, Davis.	Online
2022	Microsoft Azure Databricks for Data Engineering , Microsoft.	Online
2022	Diploma certificate in Artificial Intelligence , Sergio Arboleda University.	Online

Languages

English	Professional Working Proficiency. Currently preparing for the IELTS/TOEFL certification.
French	A1.2. Currently studying intensive courses at Alliance Française in Bogotá, Colombia